

Swaraj Borse

Walchand College of Engineering, Sangli

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Summary

Software Engineer experienced in building scalable backend systems and RESTful APIs with Java, Spring Boot, PostgreSQL, Redis, and Apache Kafka. Skilled in distributed systems, microservices, event-driven architecture, and Docker-based deployment, with strong Data Structures and Algorithms fundamentals (250+ LeetCode problems solved). Proven record of improving system performance and throughput in production-grade projects.

Technical Skills

Languages: Java, SQL, JavaScript
Backend: Spring Boot, Spring Data JPA, REST APIs, Apache Kafka, JWT, OAuth 2.0
Frontend: React.js, HTML, CSS
Database: PostgreSQL, MySQL, Redis
Cloud: AWS S3, Render
Testing: JUnit 5, Mockito, Testcontainers
Tools: Docker, Docker Compose, FFmpeg, Git, GitHub, Postman, VS Code, IntelliJ IDEA, Maven
Course Work: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

Projects

API Rate Limiter

[\[Github\]](#)

- Architected a **production-grade API Rate Limiter** using the **Token Bucket algorithm**, executing rate-limiting logic atomically inside **Redis Lua scripts** to enforce per-client limits and eliminate race conditions under **1,000+ concurrent requests**.
- Designed a **hybrid persistence architecture**, storing client configuration in **PostgreSQL** and mutable bucket state in **Redis Hashes**, reducing database writes while enabling **O(1)** bucket updates with a **single Redis EVAL** per request.
- Improved performance by migrating bucket state from PostgreSQL to Redis, increasing throughput by **93% (365 → 704 req/sec)** and reducing average latency by **52% (255 ms → 123 ms)**, while lowering **P95 latency to 257 ms**.
- Developed a **custom multithreaded benchmarking framework** to validate deterministic concurrency behavior, consistently achieving **100 allowed / 900 rejected requests, zero race conditions, and zero HTTP errors** under concurrent load.
- Built configurable per-client REST APIs with request validation, centralized exception handling, Dockerized deployment using **Docker Compose**, environment-based configuration, and comprehensive testing with **JUnit 5, Mockito, and Testcontainers**.

Technologies Used: Java 17 | Spring Boot | PostgreSQL | Redis | Lua Scripts | Docker | JUnit 5 | Mockito | Testcontainers

Video Processing Queue

[\[Github\]](#)

- Architected an **asynchronous video-processing system** using Kafka with **3 partitions**, decoupling API uploads from resource-intensive background FFmpeg processing.
- Engineered an **event-driven processing pipeline** coordinating Kafka consumers, PostgreSQL job state, S3 object storage, isolated workspaces, and background FFmpeg workers.
- Implemented a durable **4-state job lifecycle** with PostgreSQL, tracking processing status, progress, timestamps, retry metadata, errors, and indexed job/video identifiers.
- Hardened **FFmpeg execution** with a **30-minute timeout**, exit-code validation, output verification, interruption handling, and workspace cleanup; measured **~0.23s** processing time for the test video.

Technologies Used: Java 21 | Spring Boot | Apache Kafka | PostgreSQL | Spring Data JPA | Hibernate | S3 | FFmpeg | FFprobe | Docker | Docker Compose | Maven

Education

Bachelor of Technology (B.Tech) in Information Technology

CGPA: 8.65 2024 – 2028

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Achievements

- LeetCode:** Solved **300+ DSA problems**, reaching a peak rating of **1500+**.
- CodeChef:** Achieved a peak rating of **1400+ (2 Star)**.
- Solved **500+ competitive programming problems** across various difficulty levels.
- TechSprint Hackathon Finalist** – Google Developer Student Clubs (GDSC), WCE.
- Hacktoberfest:** Earned **Hacktoberfest 2025 Super Contributor** recognition for **8 merged pull requests** across open-source repositories during the annual global event.

Positions of Responsibility

Cyber Security Coordinator, WCE MLSC

2025 – 2026

- Organized and presented a Cloud Computing workshop for **200+ students**, simplifying core cloud concepts and real-world use cases.

Assistant Coordinator, TEDxWCE Sangli

2024 – 2025

- Directed logistics, speaker coordination, and execution for a flagship **TEDx** event.